



A
LAB FILE REPORT
ON
INTERNET AND WEB TECHNOLOGIES (28251107)



Submitted by:
Karan Jigyasi
BTAM25O1036
AIML

Submitted to:
Ms. Archana Acharya,
Assistant professor
Dr. Sumit Dhariwal,
Assistant professor
Dr. Vibha Tiwari,
Assistant professor

Centre for Artificial Intelligence
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Experiment – 1

Aim: Install Cisco packet tracer

Cisco Packet Tracer is a powerful network simulation tool developed by Cisco systems. It allows users, particularly students and network professionals to design, configure, simulate and troubleshoot computer networks without the need for physical hardware.

Key features of Cisco packet tracer:

Network Design & Simulation: Create and simulate complex network topologies with various Cisco devices.

Protocol Support: Simulates many protocols like RIP, OSPF, EIGRP, VLANs, and more.

Learning Tool: Widely used in Cisco Networking Academy for hands-on learning and certification prep. **Real-Time & Simulation Modes:** Visualize network traffic and troubleshoot issues in real-time.

Cross-Platform: Available on Windows, macOS, and Linux.

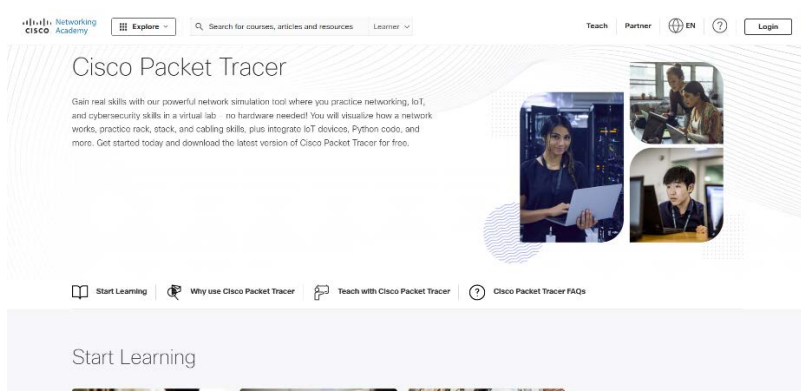
Multi-User Collaboration: Supports collaborative projects between multiple users.

Cost-Effective: Allows network design without physical hardware.

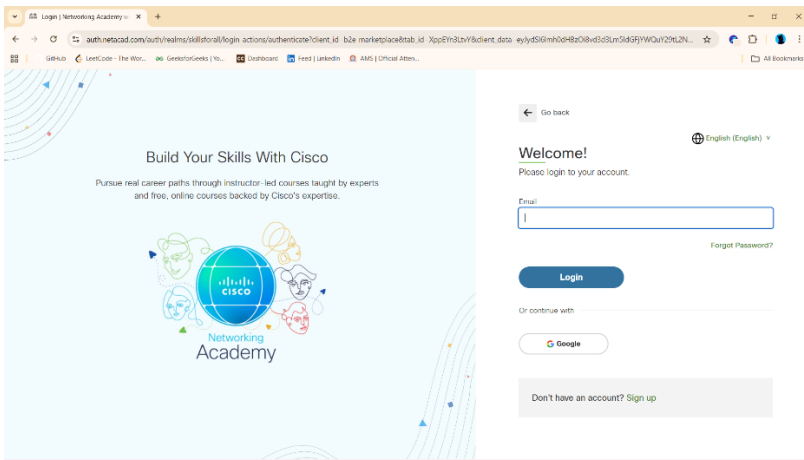
STEPS TO DOWNLOAD CISCO PACKET TRACER

Step 1: Search on chrome or any other browser Cisco packet tracer.

Step 2: Click on the first website, then the new window opens, and it will ask you to sign up and make account



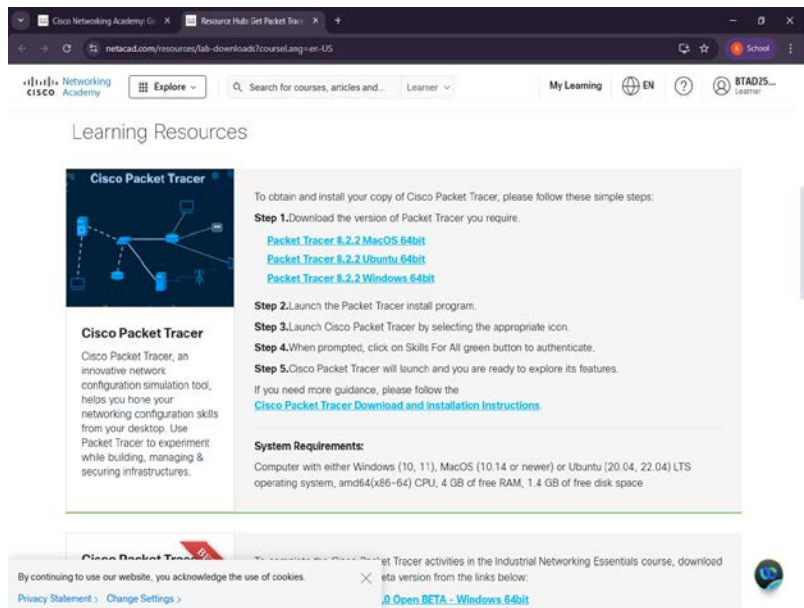
Step 3: Now login in your account.



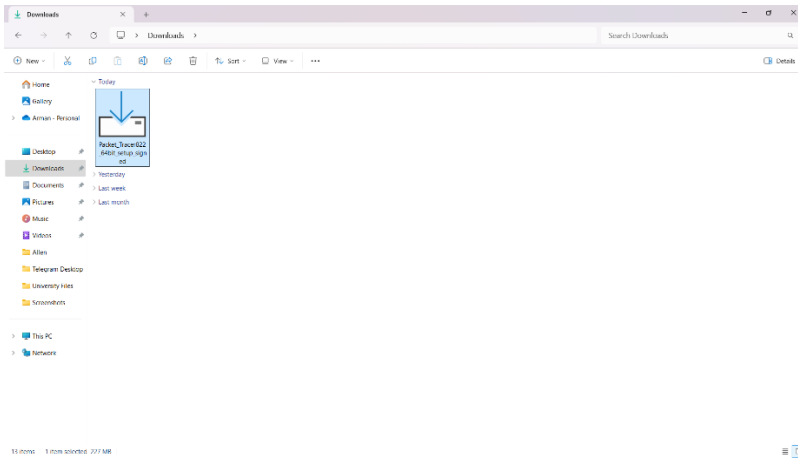
Step 4: Now click on the link to download Cisco packet tracer.



Step 5: Click on the packet tracer 8.2.2 windows 64 bit.

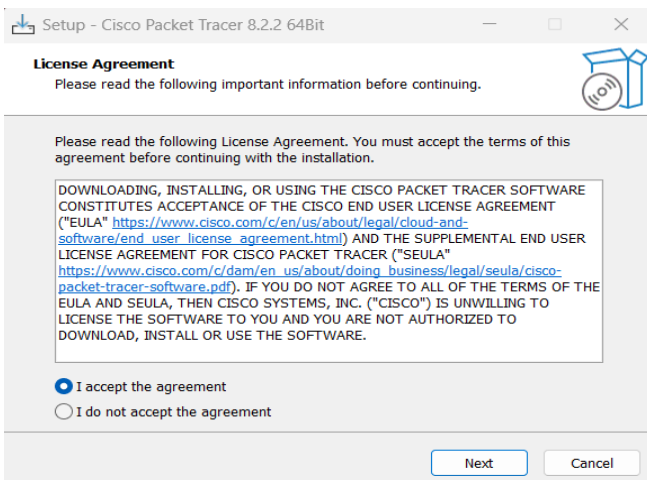


Step 6: You can see here it is downloading.



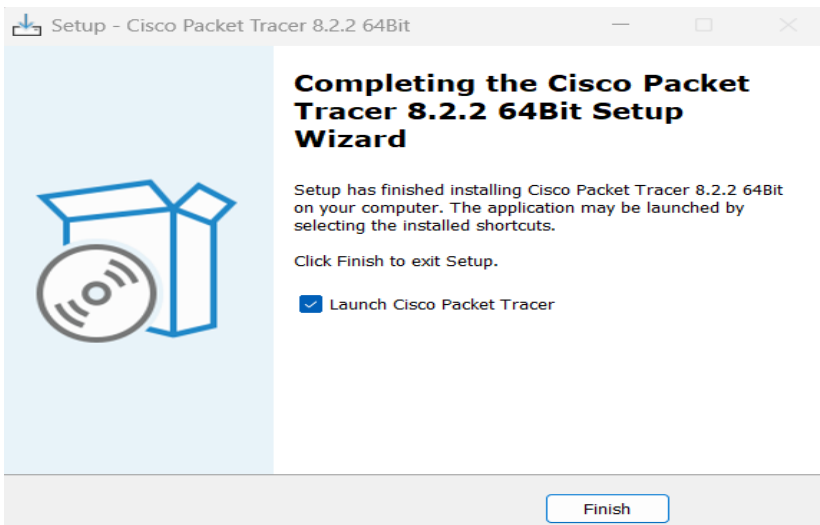
Step 7: After downloading accept the agreement and click on next.

Step 8: Again click on next.



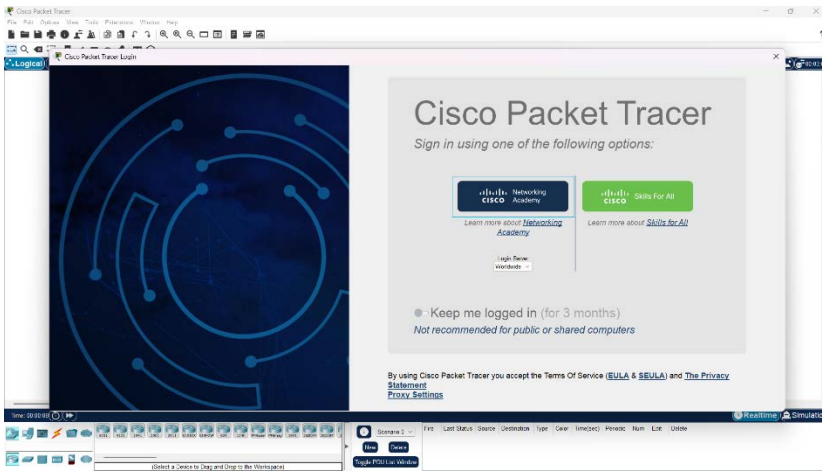
Step 9: After installing get started.

Step 10: Now click on finish.



Step 11: After this click on the green button (Cisco skills for all).

Or you can also use it offline.



Conclusion- we are successfully downloading and installing Cisco Packet Tracer by following proper precautions and the procedure given above. It is now ready for the network setup simulation.

Precautions-

1. Download Cisco Packet Tracer from the official Cisco Networking Academy Website to avoid viruses and malfunctions in software.
2. Use the correct version of Packet Tracer according to your operating system.

Experiment 2

Aim: Design a network with two PCs connected through a switch, assign appropriate IP addresses, and ensure successful communication.

Software Required: Cisco Packet Tracer

Theory:

1. **Switch** - A network switch is a device used to connect multiple devices (like PCs) on the same local network. It operates at the Data Link Layer (Layer 2) of the OSI model. It uses MAC addresses to forward data frames between devices, ensuring efficient network communication.
2. **IP** - Each device in a network must have a unique IP address for communication. For this experiment, we will assign private IP addresses from the 192.168.1.0/24 network range.
For example:
 - PC1: 192.168.1.1
 - PC2: 192.168.1.2
3. **Cisco Packet Tracer** - It is a network simulation tool used to model complex networks. It allows users to simulate the configuration of Cisco devices, like switches and routers, and understand real-world networking.

Procedure –

1. Download cisco packet tracer and open it.
2. Taking 2 PCs and 1 switch.
3. Connect the PC with a switch by using an Ethernet cable.
4. Assign IP addresses to both PC
 - a. PC1 - 192.168.1.1 [Subnet Mask - 255.255.255.0]
 - b. PC2 - 192.168.1.2 [Subnet Mask - 255.255.255.0]

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.1.1
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	

5. Open command prompt in PC1 and write “ ping 192.168.1.2”, and similarly in PC2, write. “ping 192.168.1.1” .

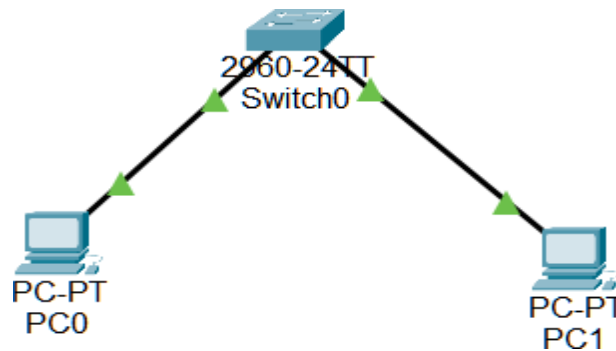
```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=10ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

6. After successful ping, transfer file from PC1 to PC2.



Conclusion: In conclusion, we designed a simple local area network (LAN) with two PCs connected through a switch. After assigning appropriate IP addresses to both PCs and configuring the network, we successfully verified communication between the devices using the ping command.

Precautions:

1. Ensure correct IP addressing.
2. Use proper cables (straight-through).
3. Check device connections to the switch.
4. Verify ping responses to avoid network issues

Experiment 3

Aim: Create a network topology that includes two subnets connected by a router. Assign IP addresses and verify connectivity between devices.

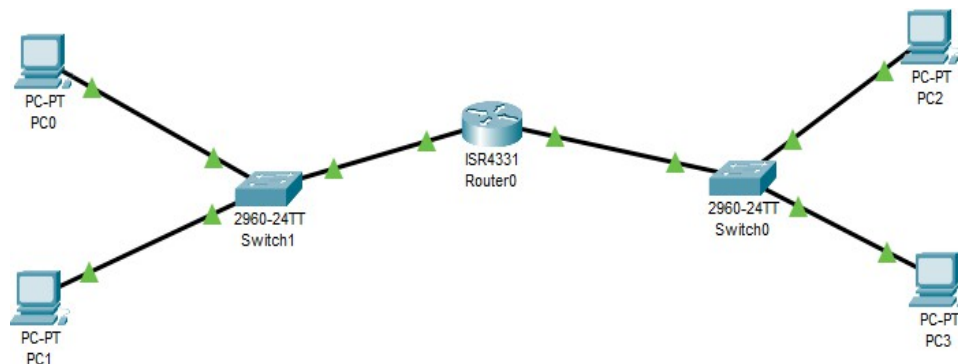
Software Required: Cisco Packet Tracer

Theory:

A **router** connects different subnets and forwards data between them using IP addresses. Each device within a subnet must have a unique IP, and the router must be configured to route traffic between the subnets. A **network topology** is the arrangement of various elements (links, nodes, and devices) in a network. In this experiment, two subnets will be connected through a router, which acts as an intermediary between them. Each **subnet** will have its own unique network address, and the router will route traffic between them. The router uses its interfaces, which are assigned **IP addresses** from each subnet, to establish communication. This setup allows devices from one subnet to communicate with devices in the other subnet, facilitating data transmission across different networks. Verification of connectivity is done using the **ping** command to ensure proper routing.

Procedure –

1. Download cisco packet tracer and open it.
2. Taking 4PCs, 2 switches and 1 router.
3. Connected PCs to switches to make a subnetwork and switches to routers with the help of cable.



4. Assigning IPs and gateways to PCs.

The screenshot shows a network configuration window titled "IP Configuration" with a blue header bar and a close button (X). The "Interface" dropdown menu is set to "FastEthernet0". Below the header, the "IP Configuration" section contains two radio buttons: "DHCP" (unselected) and "Static" (selected). The "Static" configuration fields are as follows:

Field	Value
IPv4 Address	198.162.1.2
Subnet Mask	255.255.255.0
Default Gateway	198.162.1.1
DNS Server	0.0.0.0

5. Turning on Gigabit Ethernet port and assigning IPs.

The screenshot shows a configuration window for "GigabitEthernet0/0/0". The "Port Status" is set to "On" (checked). The "Bandwidth" is set to "1000 Mbps" (selected). The "Duplex" is set to "Full Duplex" (selected). The "MAC Address" is "0060.70CE.0201". The "IP Configuration" section contains the following fields:

Field	Value
IPv4 Address	198.162.1.1
Subnet Mask	255.255.255.0

The "Tx Ring Limit" is set to "10".

6. Open the command panel and ping within the subnet and outside the subnet.

```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 198.162.1.2

Pinging 198.162.1.2 with 32 bytes of data:

Reply from 198.162.1.2: bytes=32 time<1ms TTL=127
Reply from 198.162.1.2: bytes=32 time=1ms TTL=127
Reply from 198.162.1.2: bytes=32 time<1ms TTL=127
Reply from 198.162.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 198.162.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 198.162.2.3

Pinging 198.162.2.3 with 32 bytes of data:

Reply from 198.162.2.3: bytes=32 time<1ms TTL=128
Reply from 198.162.2.3: bytes=32 time<1ms TTL=128
Reply from 198.162.2.3: bytes=32 time<1ms TTL=128
Reply from 198.162.2.3: bytes=32 time=1ms TTL=128

Ping statistics for 198.162.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

7. After successful pinging, transfer one file to one subnet to other and one PC to other within the network.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC0	PC3	ICMP		0.000	N	0	(edit)
	Successful	PC0	PC1	ICMP		0.000	N	1	(edit)

Conclusion : In this experiment we successfully designed and implemented a network topology using Cisco Packet Tracer, consisting of two subnets connected by a router. The configuration involved assigning IP addresses to devices in both subnets, configuring router interfaces to facilitate communication between the subnets, and verifying connectivity using the ping command.

Precautions:

1. Ensure correct IP addressing for all devices.
2. Use appropriate cables (straight-through) for connections.
3. Verify the router's configuration for routing between subnets.
4. Ensure that all devices and interfaces are powered on and properly connected.

Experiment 4

Aim: Construct a network with a server configured to provide FTP services. Assign IP addresses and allow PCs to upload and download files via FTP.

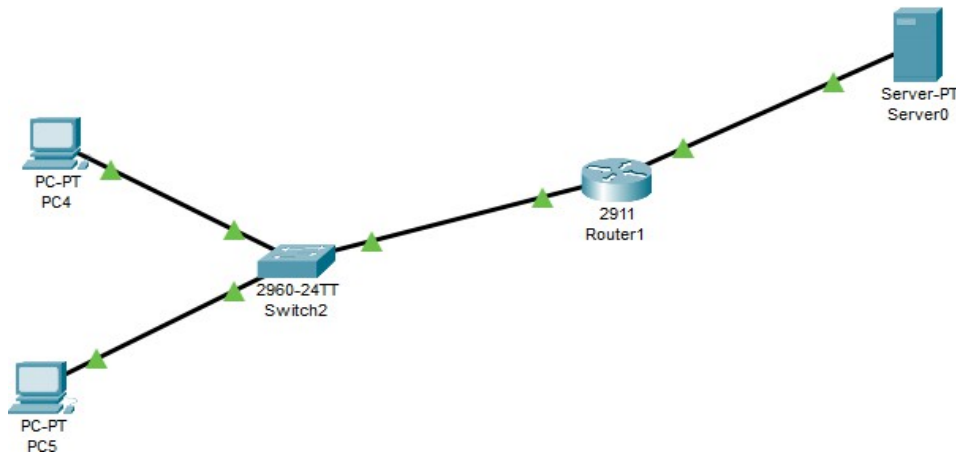
Software Required: Cisco Packet Tracer

Theory:

FTP (File Transfer Protocol) is a standard network protocol used for transferring files between a client and server. In this experiment, a server will be configured to provide FTP services, allowing connected PCs to upload and download files. A **Router** will be used to enable communication between different subnets, while a **Switch** connects the server and PCs within the same subnet, allowing data exchange. The router routes traffic between subnets, and the switch ensures efficient communication between devices within the network. Together, they create a functional network where FTP services can be accessed seamlessly.

Procedure –

1. Download cisco packet tracer and open it.
2. Taking 2 PCs, 1 switch, 1 router, 1server.
3. Connected PCs to switch, switch to router, router to server with the help of cables.



4. Assigning IPs and default gateways to PCs and servers.

The screenshot shows the 'IP Configuration' window for the 'FastEthernet0' interface. The 'Static' radio button is selected. The fields are filled with the following values:

Field	Value
IPv4 Address	198.162.10.2
Subnet Mask	255.255.255.0
Default Gateway	198.162.10.1
DNS Server	0.0.0.0

5. Turning on and assigning IPs in Gigabit Ethernet of the router.

The screenshot shows the configuration window for 'GigabitEthernet0/0'. The 'Port Status' is checked 'On'. The 'Bandwidth' is set to '1000 Mbps' and 'Duplex' is set to 'Full Duplex', both with 'Auto' selected. The 'MAC Address' is '0060.3EED.6901'. The 'IP Configuration' section shows the following values:

Field	Value
IPv4 Address	198.162.10.1
Subnet Mask	255.255.255.0

The 'Tx Ring Limit' is set to '10'.

6. Go to FTP services in the servers-
- Turning on the FTP service
 - Setup username and password
 - Allowing them important permissions.

The screenshot shows the 'Services' configuration window for 'FTP'. The 'Service' is checked 'On'. The 'User Setup' section includes fields for 'Username' and 'Password', and checkboxes for 'Write', 'Read', 'Delete', 'Rename', and 'List'. A table lists the configured users:

	Username	Password	Permission
1	cisco	cisco	RWDNL
2	kartikey	7440	RWNL

Buttons for 'Add', 'Save', and 'Remove' are located to the right of the table.

7. Open the command panel of the server, ping with both PCs one by one.

```
C:\>ping 198.162.10.2

Pinging 198.162.10.2 with 32 bytes of data:

Reply from 198.162.10.2: bytes=32 time<1ms TTL=127
Reply from 198.162.10.2: bytes=32 time<1ms TTL=127
Reply from 198.162.10.2: bytes=32 time<1ms TTL=127
Reply from 198.162.10.2: bytes=32 time<1ms TTL=127

Ping statistics for 198.162.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 198.162.10.3

Pinging 198.162.10.3 with 32 bytes of data:

Reply from 198.162.10.3: bytes=32 time<1ms TTL=127
Reply from 198.162.10.3: bytes=32 time<1ms TTL=127
Reply from 198.162.10.3: bytes=32 time<1ms TTL=127
Reply from 198.162.10.3: bytes=32 time<1ms TTL=127

Ping statistics for 198.162.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

8. After successful ping, upload and download files from PC to FTP server.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	Server0	PC4	ICMP		0.000	N	0	(edit)
	Successful	PC5	Server0	ICMP		0.000	N	1	(edit)

Conclusion : In this experiment, we successfully established a network with a server configured to provide FTP services. After assigning appropriate IP addresses to the server and PCs. We established connections that allowed file uploads and downloads via FTP. The server's configuration and the use of client software enabled efficient file transfers between the devices.

Precautions:

1. Ensure that the correct IP addresses and subnet masks are assigned to all devices.
2. Verify that the router's interface is configured and active.
3. Ensure that the FTP service is enabled on the server.

Test basic connectivity using the ping command before attempting FTP file transfers.

Experiment 5

Aim: Design a network with an SMTP server to handle email communication. Assign IP addresses, configure email clients on PCs, and send test emails.

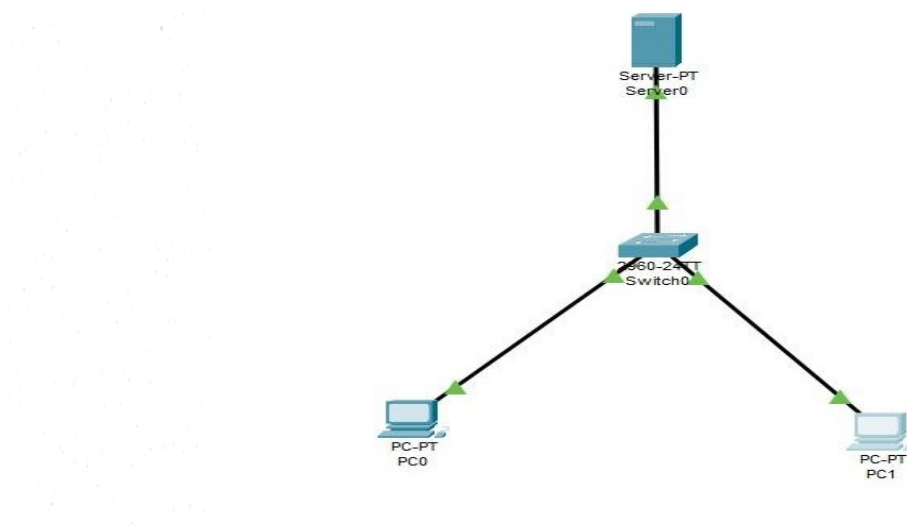
Software Required: Cisco Packet Tracer

Theory:

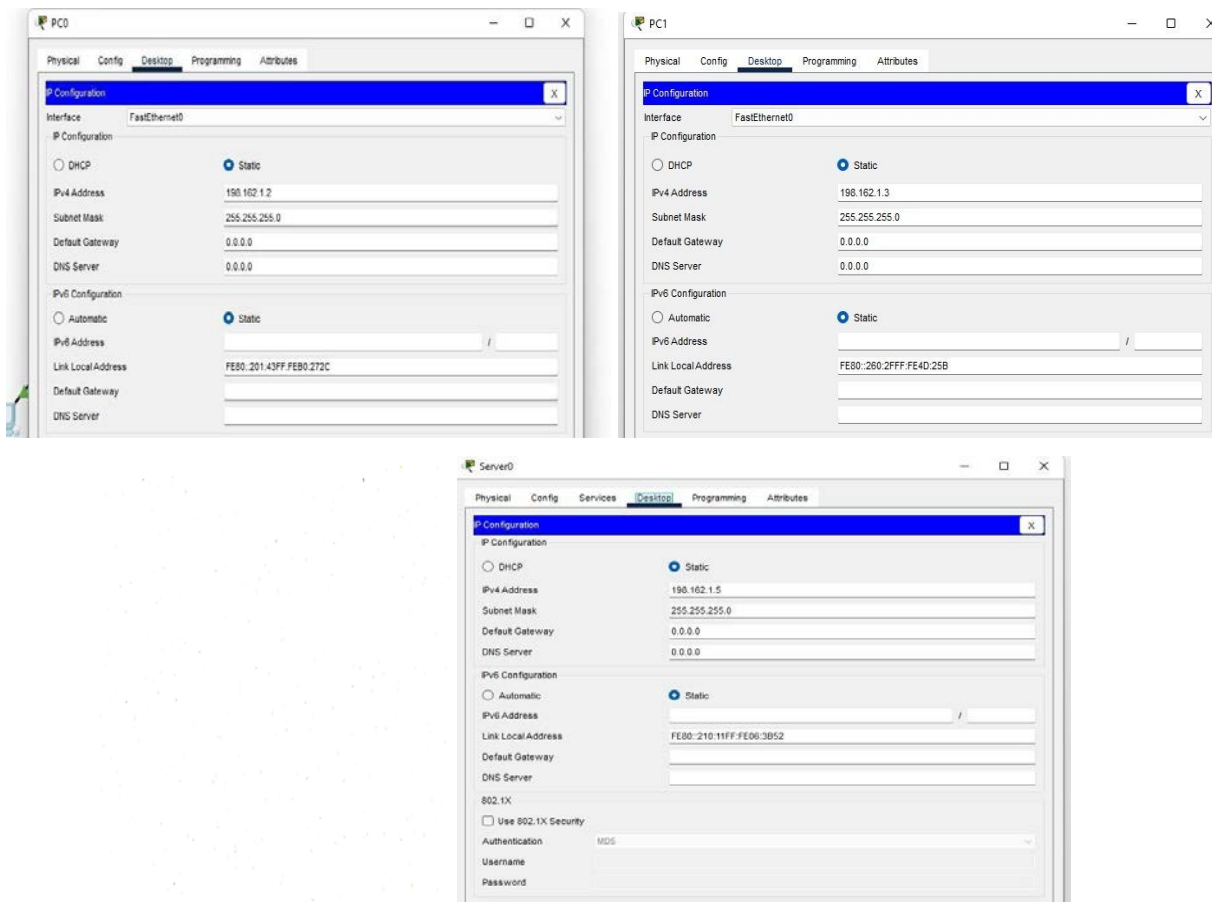
Simple Mail Transfer Protocol (SMTP) is an internet standard for sending emails across IP networks. It works on port 25 and is responsible for routing emails between mail servers and clients. In this experiment, we will configure a network with an SMTP server and use PCs as email clients to test the email sending process.

Procedure -

1. Open Cisco Packet Tracer.
2. Taking 2 PCs, 1 switch, 1 server.
3. Connecting PCs to switch, switch to server with the help of cables.

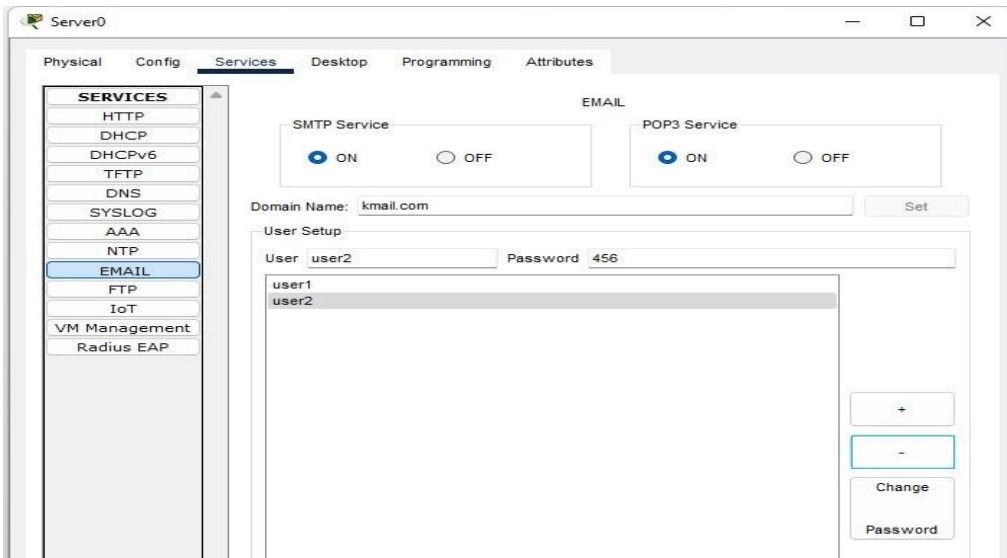


4. Assigning IP addresses to PC0, PC1 and server
 - Server - IP Address: 192.168.1.5, Subnet Mask: 255.255.255.0
 - PC0 - IP Address: 192.168.1.2, Subnet Mask: 255.255.255.0
 - PC1 - IP Address: 192.168.1.3, Subnet Mask: 255.255.255.0



5. Configuration of SMTP Server-

- Click on the server, go to the Services and select SMTP.
- Check the SMTP service is "On", if not then "On" it by click.
- Giving domain name and setup account for PCs-
 - Server - Domain name: kmail.com
 - PC0 - Email: user1@kmail.com, Password: 123
 - PC1 -Email: user2@kmail.com, Password: 456



6. Configure Email Clients on PCs:

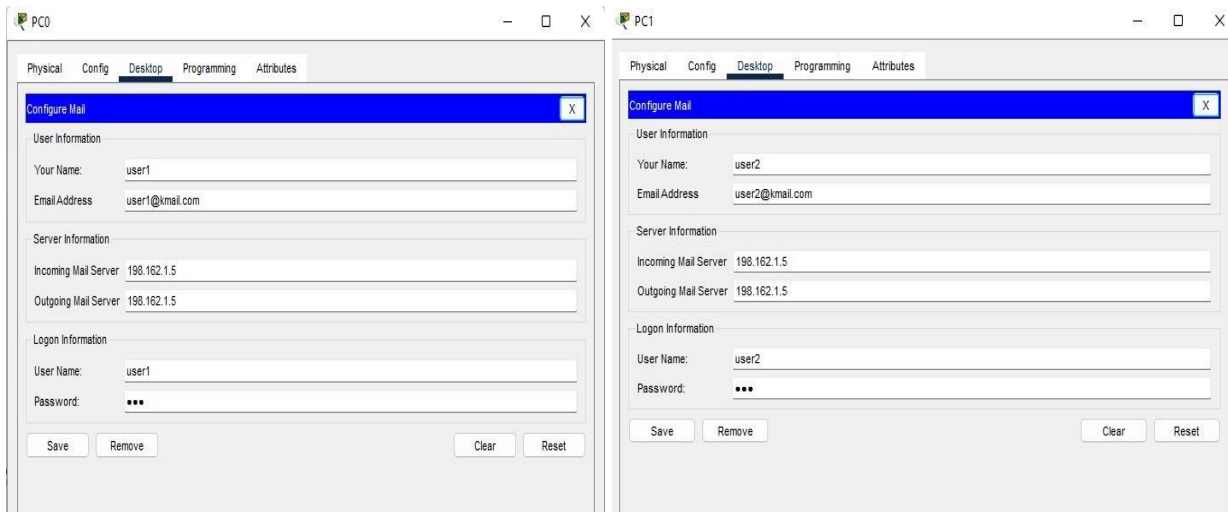
a. Open PC, go to the Desktop tab and open the Email application.

■ Configure of PC0 as follows:

- Email: user1@kmail.com
- SMTP server: 192.168.1.5
- Username: user1
- Password: 123

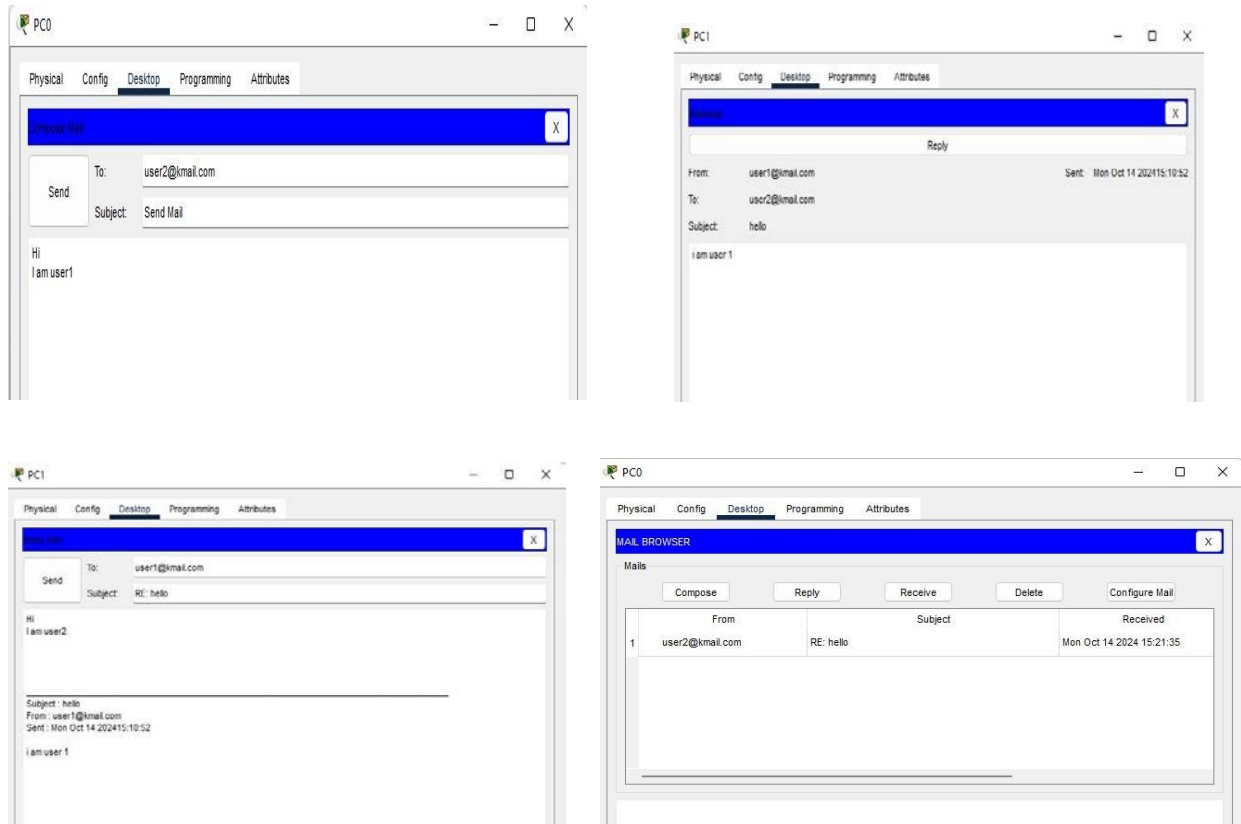
■ Configure of PC1 as follows:

- Email: user2@kmail.com
- SMTP server: 192.168.1.5
- Username: user2
- Password: 456



7. Sending and receiving email-

- From PC0, compose and send an email to user2@kmail.com.
- On PC1, open the email and receive the email from PC0.
- Response of PC1 to user1@kmail.com.
- Other side also received email.



Conclusion - We successfully established connections between SMTP server and PCs. This exercise helped us understand how to set up and configure a network for email services. Both PCs were able to send and receive emails, demonstrating proper email functionality within the network.

Precautions -

1. Verify that the SMTP service is enabled on the server.
2. Establish the proper configuration of SMTP Server, like username, password, IP address, etc.
3. After setting up email account details on PCs, you must click on the save button.

Experiment 6

Aim: Create a network with a DNS server that resolves domain names for multiple devices. Ensure that PCs can access each other and external services using domain names instead of IP addresses.

Software Required: Cisco Packet Tracer

Theory:

The Domain Name System (DNS) translates hostnames into IP addresses. In a local network:

Internal PCs can communicate using hostnames instead of IPs.

DNS servers can forward requests to external DNS servers to resolve public domain names.

Forward Lookup Zone: Maps hostnames to IP addresses.

Forwarders: Forward unresolved requests to public DNS servers.

Procedure -

Step 1: Build the Network

1. Open Cisco Packet Tracer.
2. Place devices:
 - o 1 Switch
 - o 2 PCs
 - o 1 Server (DNS)
3. Connect all devices to the switch using copper straight-through cables.

Step 2: Assign IP Addresses

Device	IP Address	Subnet Mask	Default Gateway
PC1	192.168.1.101	255.255.255.0	192.168.1.1
PC2	192.168.1.102	255.255.255.0	192.168.1.1
DNS Server	192.168.1.10	255.255.255.0	192.168.1.1

Experiment 7

Aim: Develop a network that integrates both SMTP and FTP services on a single server. Configure the PCs to access email services and file transfers.

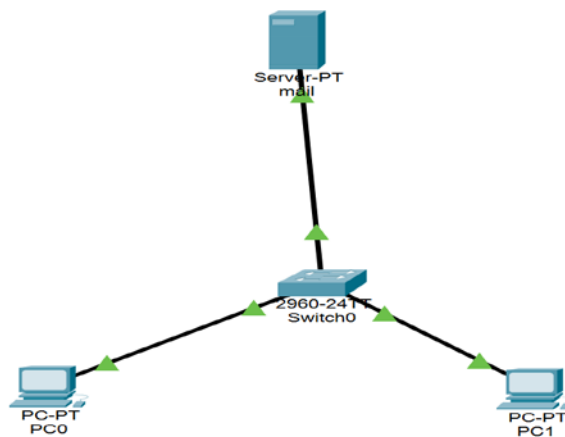
Software Required: Cisco Packet Tracer

Theory:

- SMTP (Simple Mail Transfer Protocol) sends outgoing emails.
- POP3 or IMAP retrieves incoming emails.
- FTP (File Transfer Protocol) allows uploading and downloading files between client and server.
Running both services on one server lets users communicate and share resources within the same LAN.

Procedure –

1. Design the network: Connect the server and PCs to a switch with straight-through cables.



2. Assign IPs:

Device	IP Address	Subnet Mask	DNS
Server	192.168.1.1	255.255.255.0	—
PC1	192.168.1.2	255.255.255.0	192.168.1.1
PC2	192.168.1.3	255.255.255.0	192.168.1.1

mail

Physical Config Services **Desktop** Programming Attributes

IP Configuration X

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.1

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20D:BDF:FE4:51DD

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.2

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D9:D3FF:FE82:77B8

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.3

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:97FF:FE65:4943

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

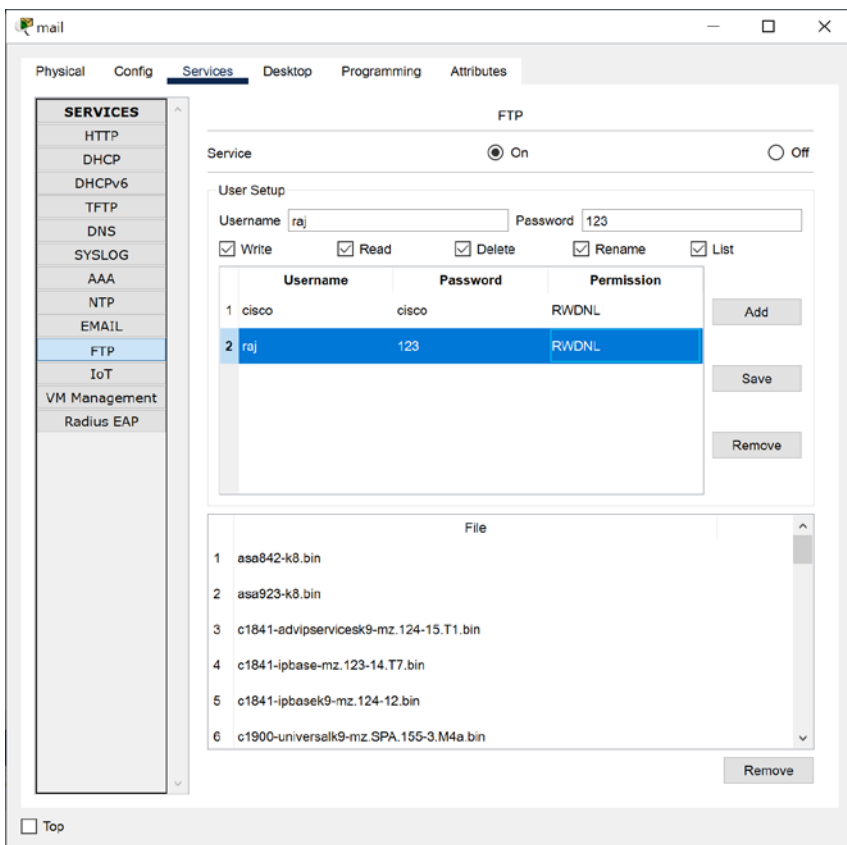
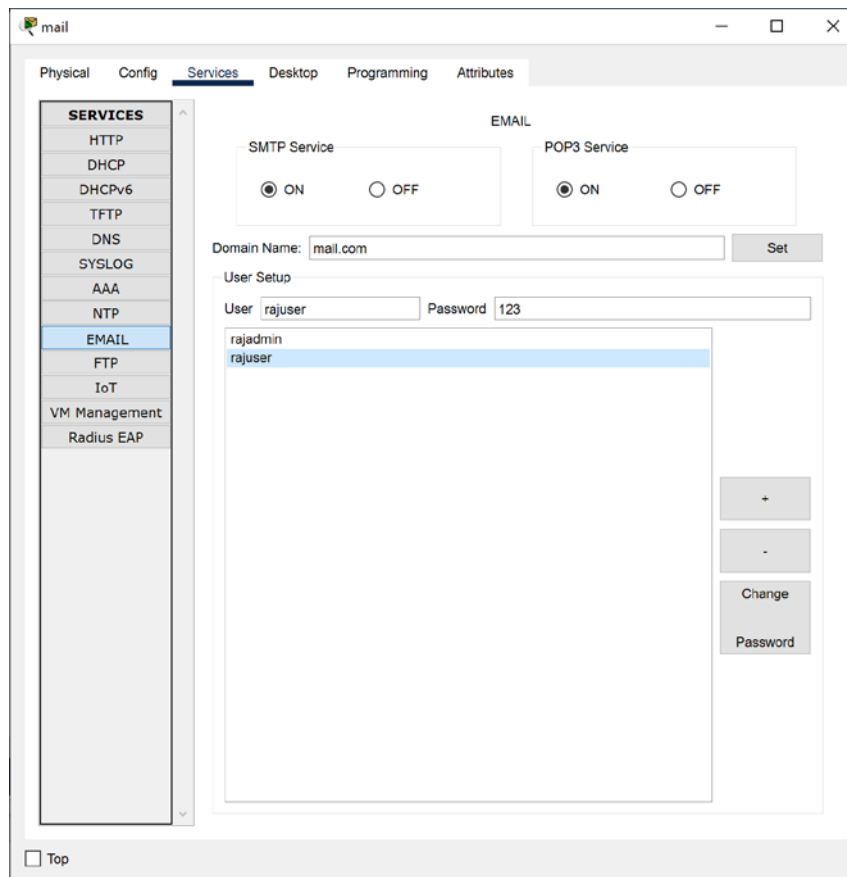
Username

Password

☐ Top

3. Configure the Server:

- Turn ON SMTP and POP3, create user accounts (e.g., *rajadmin@mail.com*, rajuser@mail.com).
- Enable FTP, add users (same usernames/passwords).



4. Configure PCs:

- In each PC → Desktop → Email:
 - Set mail server to 192.168.1.1
 - Login using created accounts.
- In each PC → Desktop → Command Prompt:
 - Test FTP: [ftp 192.168.1.1](ftp://192.168.1.1)

PC0

Physical Config **Desktop** Programming Attributes

Configure Mail X

User Information

Your Name:

Email Address:

Server Information

Incoming Mail Server:

Outgoing Mail Server:

Logon Information

User Name:

Password:

Save Remove Clear Reset

☐ Top

PC1

Physical Config **Desktop** Programming Attributes

Configure Mail X

User Information

Your Name:

Email Address:

Server Information

Incoming Mail Server:

Outgoing Mail Server:

Logon Information

User Name:

Password:

Save Remove Clear Reset

☐ Top

PC0

Physical Config **Desktop** Programming Attributes

Command Prompt X

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 192.162.1.1
Trying to connect...192.162.1.1
%Error opening ftp://192.162.1.1/ (Timed out)
.

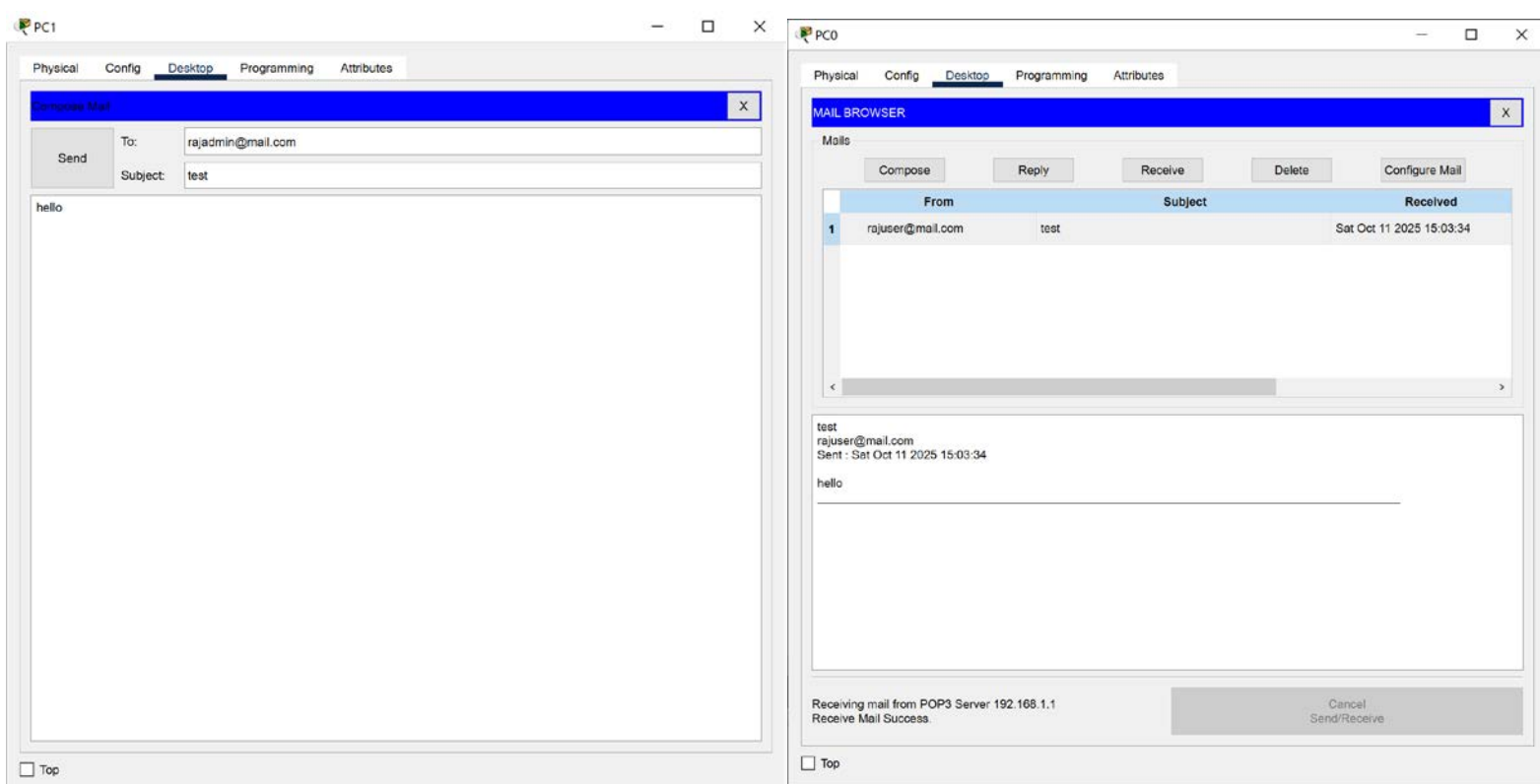
(Disconnecting from ftp server)

ftp 192.168.1.1
Trying to connect...192.168.1.1
Connected to 192.168.1.1
220- Welcome to PT Ftp server
Username:raj
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>
```

☐ Top

5. Testing:

- Send an email from PC1 to PC2 and verify receipt.



- Upload a file from PC1 to the server via FTP and download it from PC2.

Conclusion:

Both SMTP and FTP services were successfully configured on one server, allowing email communication and file transfer among PCs.

Precautions:

- Ensure correct IP and gateway settings.
- Use matching usernames/passwords on both services.
- Verify SMTP, POP3, and FTP services are switched ON.
- Check firewall settings if emails or transfers fail.

